Extended Abstract

Motivation Sketches are powerful tools for abstraction, capturing the essence of an image with sparse, intentional strokes. Yet generating sketches that are both semantically meaningful and visually coherent remains challenging. Prior methods have relied on feedforward generation or optimization-based techniques, which lack the sequential decision-making process of human sketching. We ask: can reinforcement learning better model sketching as a reward-driven, coarse-to-fine behavior?

Method We frame sketch generation as a two-stage sequential decision-making problem that mirrors how humans sketch—starting with rough structure and refining it over time. In the first stage, we train a behavior cloning agent to predict Bézier stroke endpoints directly from the input image. This agent is supervised using expert demonstrations from a large-scale photo-sketch dataset, enabling it to learn high-level spatial placement strategies. Each predicted stroke is initially rendered as a straight line, providing a coarse approximation of the target sketch.

In the second stage, we introduce a reinforcement learning agent that refines the curvature of each stroke by predicting control points for the associated Bézier curve. This agent is trained using the REINFORCE algorithm and receives a composite reward signal comprising three components: (1) semantic alignment, measured via CLIP similarity between the rendered sketch and the input image; (2) geometric fidelity, computed as silhouette overlap with edge maps from the input; and (3) aesthetic quality, captured through penalties on high curvature and irregular stroke shapes. The refinement stage leverages a differentiable rendering engine to rasterize the updated strokes and compute gradients for the reward function.

By separating placement and refinement, our method enables more stable training and modularity. The placement agent provides a strong prior that constrains the action space for the RL agent, which then focuses on low-level refinements that improve sketch quality in a reward-driven manner.

Implementation We implement our approach in PyTorch and use a differentiable renderer to convert vector strokes to raster images. The stroke placement agent is trained on 14,000 samples from the ControlSketch dataset, mimicking expert demonstrations. The refinement agent is trained via reinforcement learning, receiving feedback from CLIP similarity, silhouette difference, and aesthetic regularizers. All training is conducted on a machine with an NVIDIA RTX 4090 GPU.

Results Our method achieves strong semantic alignment and perceptual recognition. Quantitatively, it reaches 72% CLIP similarity between generated sketches and input images. Qualitatively, a user study shows 88% CLIP alignment between user-provided labels and ground-truth categories. Generated sketches are sparse yet recognizable, covering a wide range of visual categories.

Discussion Our results demonstrate that decomposing sketch generation into placement and refinement is both effective and interpretable. The behavior cloning agent successfully predicts spatially meaningful stroke locations, while the RL-based refinement agent adds curvature that improves both semantic alignment and visual appeal. This mirrors the human sketching process, where artists often begin with rough outlines and iteratively refine details.

Importantly, our two-stage formulation resolves several training challenges observed in earlier unified approaches. In our initial experiments, training a single agent to jointly predict both endpoints and control points led to unstable optimization and poor convergence. By narrowing the scope of each agent's task and providing structural priors from supervised demonstrations, we significantly improve learning stability and output quality.

While effective on simple single object inputs, the method struggles with cluttered compositions and fine-grained details. This suggests the need for hierarchical reasoning and more precise rewards. Future work may explore part-aware objectives or human-in-the-loop guidance for finer control.

Conclusion By modeling sketch generation as a sequential, reward-driven process, our framework aligns AI drawing behavior with human sketching strategies. The combination of supervised stroke placement and reinforcement-based refinement enables interpretable, goal-directed sketches. This approach advances controllable sketch synthesis and opens the door to interactive sketch-based tools.

Drawn to Reward: RL for Semantic Sketching

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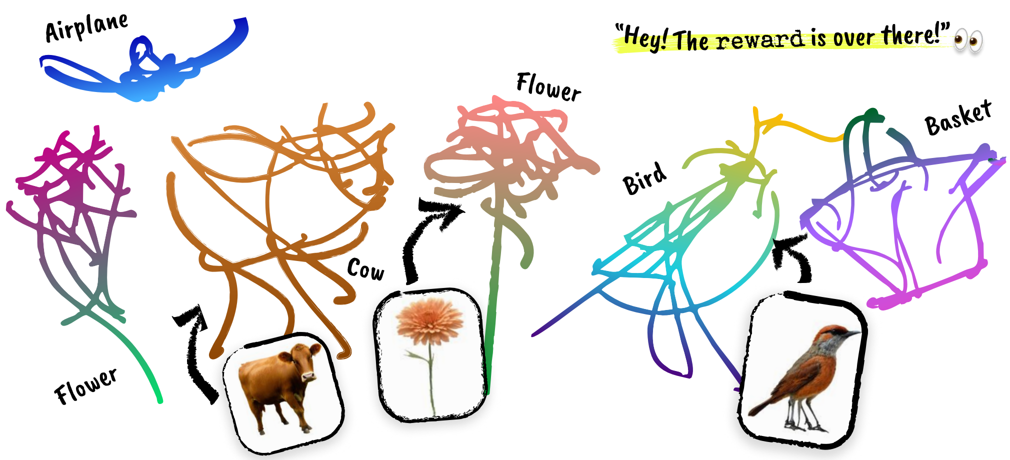
Abstract

We propose a reinforcement learning framework for generating semantically meaningful and visually coherent vector sketches from input images. Inspired by the human sketching process, our method decomposes the task into two stages: a behavior cloning agent predicts where to place strokes, and a REINFORCE agent refines their shape. We design a multi-part reward function combining CLIP-based semantic feedback, silhouette-based geometric alignment, and aesthetic regularization. Trained on a large photo-sketch dataset, our two-stage agent produces sketches that achieve 72% CLIP alignment and 88% user identification success. Our results highlight the potential of structured, reward-driven policies for controllable and interpretable sketch generation.



1 Introduction

Sketching is a powerful and intuitive form of visual communication, widely used across domains from design ideation to educational diagrams. Unlike photorealistic imagery, sketches capture the structural essence of a concept with just a few lines, enabling fast iteration, abstraction, and semantic clarity. However, despite their simplicity, generating human-like sketches remains a challenging problem in computer vision and graphics due to the need for both semantic understanding and fine-grained spatial control.

Recent progress in generative modeling has led to impressive sketch synthesis tools. Models like CLIPasso optimize strokes to match the semantics of an input image using CLIP-based guidance, while diffusion-based methods like DiffSketcher have shown promise for vector sketch generation.

Yet, most existing approaches operate in a one-shot or feedforward fashion, lacking the sequential, intentional process that underlies human sketching. This motivates the question: can reinforcement learning better model the act of sketching as a step-by-step, reward-driven behavior?

In this work, we propose a reinforcement learning framework for sketch generation that mimics how people draw: starting with coarse structure and refining with detail. Our agent receives an input image and outputs a sequence of vector strokes, each represented as a Bézier curve. To manage the complexity of this task, we decompose it into two stages. The first agent places stroke endpoints using behavior cloning on a curated sketch dataset, and the second agent refines stroke shapes via REINFORCE using a combination of semantic, geometric, and aesthetic rewards.

This two-stage approach reflects the coarse-to-fine nature of real-world sketching and improves training stability compared to a monolithic policy. We leverage CLIP for semantic feedback, edge alignment for geometric fidelity, and smoothness-based regularizers to guide refinement. Our method generates clean, semantically faithful sketches that can generalize to new input images without per-image optimization.

Through quantitative analysis and user studies, we demonstrate that our approach achieves strong alignment with input content. The proposed method bridges semantic-level intent and stroke-level execution, providing a foundation for interactive, interpretable sketch generation systems. Ultimately, this work contributes to the broader goal of aligning AI drawing behaviors with human visual reasoning and creative workflows.

2 Related Work

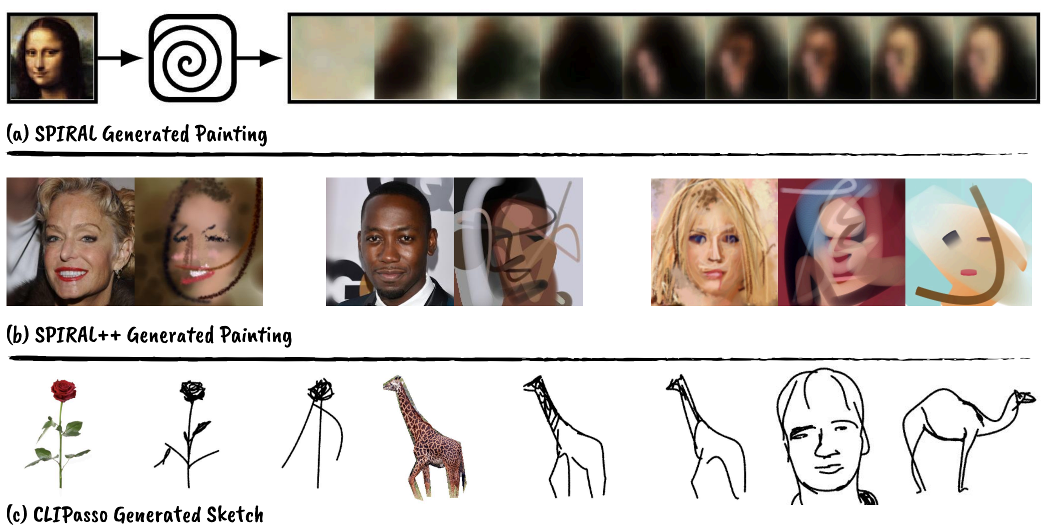


Figure 1: **Prior Work Overview.** SPIRAL and SPIRAL++ use reinforcement learning to sequentially render full-color paintings via adversarial training. CLIPasso instead directly optimizes Bézier curve strokes for semantic alignment using CLIP guidance. Our work draws inspiration from both: we adopt RL to model sequential stroke prediction, but use CLIP-based and geometric feedback for guiding abstract vector sketches.

Converting a natural image into an abstract line drawing—a sketch that is sparse yet captures the nuances of the image—has long been an interesting and important topic in computer vision and graphics. In recent years, active efforts have emerged as image synthesis techniques advanced with the rise of generative models. Prior methods have leveraged RNNs (Ha and Eck, 2017), GANs (Chan et al., 2022), optimization (Vinker et al., 2022), and more recently diffusion models (Wang et al., [n. d.]; Xing et al., 2023). These approaches generate an entire sketch, either as rasterized pixel images or structured vector representations (e.g., Bézier curves). However, when humans draw sketches, we work stroke by stroke—each decision sequentially building upon the previous ones—which naturally suggests reinforcement learning as an intuitive choice for modeling this process.

2.1 Most relevant work: SPIRAL, SPIRAL++, CLIPasso

SPIRAL and SPIRAL++ (Ganin et al., 2018; Mellor et al., 2019) train an RL agent to sequentially paint realistic images using a learned discriminator as a reward signal. CLIPasso (Vinker et al., 2022) instead uses CLIP as a pretrained reward model and directly optimizes vector stroke parameters to create semantically faithful sketches. While SPIRAL++ focuses on complex natural scenes and high-dimensional RGB outputs, CLIPasso operates in a low-dimensional vector sketch space, similar to our setting. We also take inspiration from recent CLIP-guided generative methods that use score distillation or CLIP similarity as reward signals in creative domains, though these typically use gradient descent rather than RL.

2.2 SPIRAL++ Method

Here we provide more details on the technical approach employed by the SPIRAL series work. The SPIRAL++ method builds heavily on top of the previous SPIRAL (Ganin et al., 2018) work. SPIRAL is a framework where an agent learns to generate images by sequentially issuing drawing commands (a "program") to a black-box renderer, like a painting simulator. The agent is trained via reinforcement learning to maximize rewards given by a discriminator network, which is adversarially trained to distinguish between real images and those produced by the agent-renderer pipeline. Crucially, the agent receives its reward only at the end of an episode, based on the final rendered image, making the task nontrivial since it must learn a sequence of actions that collectively fool the discriminator. Building on SPIRAL, SPIRAL++ introduces several key improvements: (1) spectral normalization to stabilize discriminator training, (2) temporal credit assignment to provide step-by-step reward signals, rather than only at the end, (3) a complement discriminator that forces the agent to match higher-level semantics instead of relying on pixel-wise similarities, and (4) population-based training where multiple agents share a discriminator, promoting diverse strategies and improved robustness. Together, these changes enable SPIRAL++ agents to create outputs ranging from abstract sketches to highly realistic images without supervision.

2.3 Existing Work Limitations

While SPIRAL++ demonstrates how RL can be used to generate realistic images, it is not suited for sparse, abstract, black-and-white sketches. Its reliance on RGB renderers and GAN-based discriminators makes it ill-equipped to handle vector sketches or compositional line drawings. It also lacks the ability to process text or image prompts directly—whereas our agent leverages CLIP to align with arbitrary prompts. CLIPasso, on the other hand, produces compelling sketches but requires a separate optimization process for each target and does not learn a generalizable drawing policy. It is not interactive, sequential, or reusable. Our work addresses these limitations by training a policy that generates vector sketches in a stroke-by-stroke manner conditioned on high-level prompts, enabling generalization and reuse while maintaining semantic alignment.

3 Method

3.1 Problem Formulation

We aim to train a reinforcement learning (RL) agent capable of synthesizing vector sketches from input RGB images. Each sketch is represented as a sequence of Bézier curves, parameterized by endpoints and control points. A differentiable rendering engine is used to convert these vector representations into rasterized images for training and evaluation. The overall goal is to maximize the semantic and visual alignment between the generated sketch and a target image.

3.2 Initial Unified Approach

Our initial approach framed the sketch generation task as a unified reinforcement learning problem, wherein a single agent was trained end-to-end to predict both stroke endpoints and control points in sequence. The agent directly generated complete Bézier curves given an input image, aiming to jointly learn where to place strokes and how to refine their shapes. However, we found this joint formulation to be unstable and sample-inefficient, primarily due to the compounded difficulty of

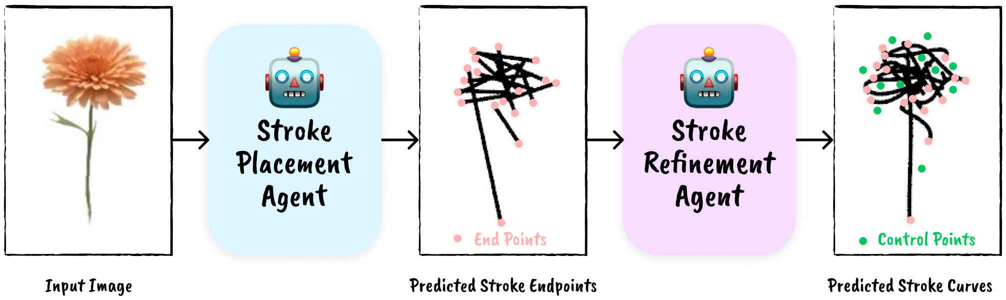


Figure 2: **Overview of our two-stage sketch generation pipeline.** Given an input image, a behavior cloning agent first predicts the endpoints of Bézier strokes (Stage 1). At this stage, we render the sketch as a sequence of straight lines, since the strokes have no curvature without control points. In Stage 2, a separate reinforcement learning agent refines each stroke by predicting control points, introducing curvature to better match the structure and semantics of the target image.

optimizing both high-level semantic alignment and low-level geometric precision within a single policy. Learning failed to converge reliably, motivating our shift to a decomposed two-stage approach.

3.3 Two-Stage Decomposition

Human artists typically sketch in a ***coarse-to-fine manner***. They begin by laying down key structural elements—such as the general outline or boundaries of an object—and incrementally refine the sketch with details, shading, and expressive curves. This process often involves first deciding on the endpoints of lines or shapes, and then refining how those lines curve and connect. Artists intuitively balance semantic intent with visual aesthetics, adjusting stroke curvature to emphasize form and flow. Inspired by this natural decomposition, we structure our model to first determine stroke placement (endpoints) and then refine each stroke's curvature (control points), enabling interpretable and modular sketch construction.

Figure 3: **Human sketching is inherently coarse-to-fine across styles and subject matter.** Whether drawing shaded studies or colorful compositions, artists begin with rough lines and iteratively refine with structure and nuance. This motivates our stroke-level decomposition: each sketch line is composed of two endpoints and a set of control points that shape its curvature.

To reflect the natural structure of the artistic process, we decompose the problem into two stages: stroke placement and stroke refinement.

- **Stroke Placement:** Determine the start and end coordinates of each Bézier curve stroke.
- **Stroke Refinement:** Adjust the control points to refine the curvature of each stroke.

This decoupling addresses the inherent difficulty of jointly optimizing both stroke location and curvature, which we empirically found to be unstable and sample-inefficient.

3.4 Stage 1: Stroke Placement via Behavior Cloning

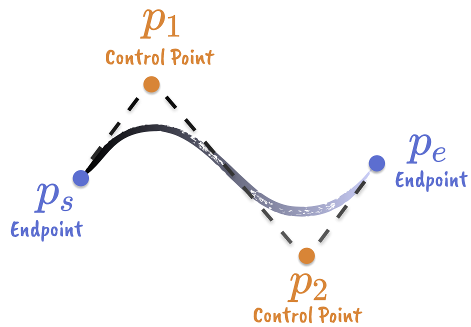


Figure 4: **Representation of a stroke.** Each stroke is defined as a cubic Bézier curve with two endpoints p_s and p_e, and two control points p_1 and p_2. The stroke placement agent predicts the endpoints, while the refinement agent adjusts the control points to shape the curvature.

We formulate the stroke placement task as a supervised learning problem, using behavior cloning. The agent is trained to mimic expert demonstrations derived from a curated dataset of photo-sketch pairs. These sketches are generated using CLIPasso, a model that produces semantically faithful sketches from photos. Each expert trajectory consists of stroke endpoints extracted from CLIPasso-generated sketches. The stroke placement agent is trained to minimize the discrepancy between its predicted stroke locations and those of the expert.

Let x_i denote the input image and y_i denote the ground-truth set of stroke endpoints. The behavior cloning agent π_θ is optimized via supervised loss:

$$L_{BC}(\theta) = \sigma_i \|\pi_\theta(x_i) - y_i\|_2^2$$

3.5 Stage 2: Stroke Refinement via Reinforcement Learning

The stroke refinement task is modeled as a Markov Decision Process (MDP), where the agent sequentially predicts control points to bend the strokes. We employ the REINFORCE algorithm to optimize the agent with respect to a reward function that combines semantic and geometric feedback. The reward r consists of three components:

- **Semantic Reward** (r_{CLIP}): Measures similarity between the generated sketch and the target image using CLIP image embeddings. CLIP (Contrastive Language–Image Pretraining) (Radford et al., 2021) is a vision-language model trained to align textual descriptions with visual content. Given an image and a text prompt, CLIP encodes both into a shared embedding space and measures their cosine similarity. In our setup, we use CLIP's image encoder to extract semantic features from both the target photo and the generated sketch, computing their similarity to guide the RL agent. This encourages sketches to preserve high-level semantics from the reference image, even when composed of sparse lines.
- **Geometric Reward** ($r_{\text{silhouette}}$): Penalizes the discrepancy between the silhouette of the rendered sketch and that of a target image.
- **Aesthetic Regularization** ($r_{\text{aesthetic}}$): Encourages smoothness and simplicity of the Bézier curves to improve visual appeal.

The total reward is given by the following sum:

$$r = \lambda_1 r_{\text{CLIP}} + \lambda_2 r_{\text{silhouette}} + \lambda_3 r_{\text{aesthetic}}$$

where $\lambda_1, \lambda_2, \lambda_3$ are weighting coefficients empirically selected through experimentation.

We train the control point refinement agent with REINFORCE gradient:

$$\nabla J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \left(\sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(a_{i,t}|s_{i,t}) \right) \left(\sum_{t=1}^{T} r(s_{i,t}, a_{i,t}) \right)$$

and we update the policy by $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$.

4 Experimental Setup

We implement our method using PyTorch and conduct all experiments on a machine equipped with an NVIDIA RTX 4090 GPU. For training the stroke placement agent, we utilize the ControlSketch dataset (Arar et al., 2025), a large-scale collection of photo-sketch pairs. Specifically, we leverage 14,000 training examples from this dataset to supervise the behavior cloning agent.

5 Results

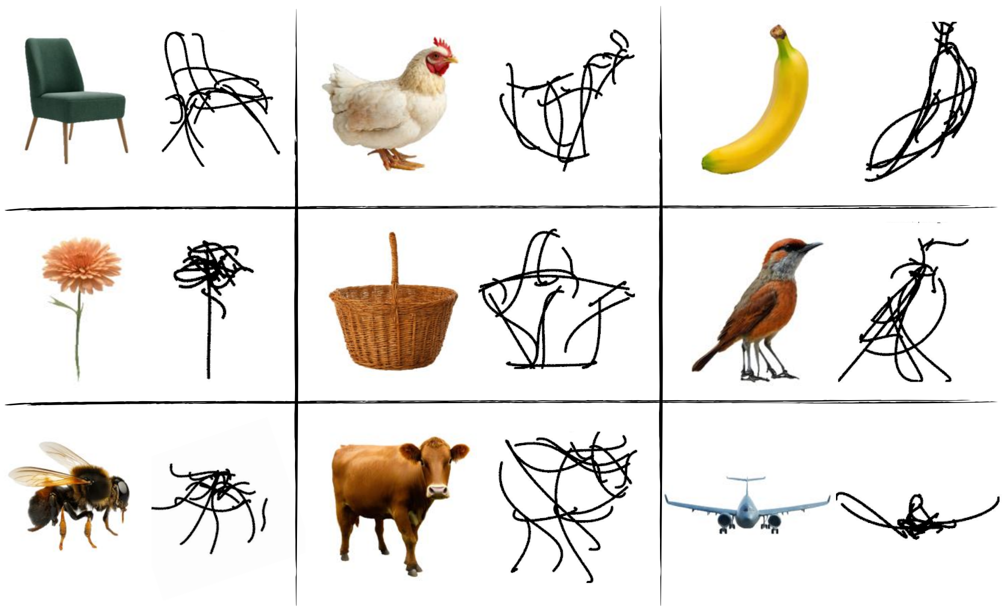


Figure 5: **Visualization of results.** Each pair shows a reference image (left) and the corresponding sketch generated by our method (right). Despite the sparse and abstract nature of the strokes, the sketches preserve the overall structure and semantic identity of the input objects. Results span a diverse range of categories, from animals and plants to furniture and tools, highlighting the model's generalization capability across visual domains.

We evaluate our method both quantitatively, using CLIP-based semantic similarity metrics, and qualitatively, through user studies assessing perceptual alignment.

5.1 Quantitative Evaluation

To assess semantic fidelity, we compute CLIP similarity between the generated sketch and the corresponding input image. Our full two-stage method—comprising a behavior cloning agent for stroke placement and a REINFORCE agent for control point refinement—achieves a **CLIP alignment score of 72%**. This number highlights the importance of task decomposition and the inclusion of multiple reward signals, leading to sketches that are both interpretable and faithful to the target.

5.2 Qualitative Evaluation

In our user study, participants were shown a generated sketch and asked to provide a label describing the object they perceived. To evaluate perceptual alignment, we computed CLIP similarity between the user-provided label and the ground-truth label associated with the original image (two text labels). Rather than relying on exact label matching, this approach accounts for the inherent ambiguity in naming objects and offers a more robust measure of conceptual alignment. The two-stage agent achieved an average **CLIP similarity of 88% between user-provided labels and ground-truth labels**, indicating that the generated sketches were conceptually aligned with the intended objects and perceptually recognizable to human viewers.

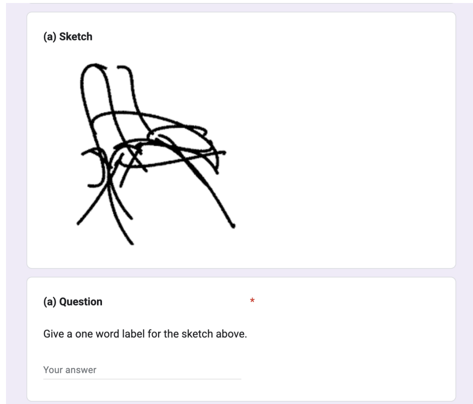


Figure 6: **User study google form.**

6 Discussion

Our results demonstrate that reinforcement learning can be effectively applied to generate semantically meaningful and visually coherent vector sketches, particularly when the sketching process is decomposed into distinct stages of stroke placement and refinement. The two-stage formulation not only mirrors how humans sketch—coarse-to-fine with iterative refinements—but also empirically improves learning stability and output quality compared to a monolithic end-to-end agent.

The behavior cloning agent, trained on a large-scale photo-sketch dataset, successfully learns to initialize stroke endpoints in a spatially meaningful way. This initialization is critical, as it constrains the search space for the downstream refinement agent and provides a strong prior for semantic structure. Meanwhile, the refinement agent benefits from a carefully designed reward function that balances high-level semantic alignment (via CLIP), low-level geometric fidelity (via silhouette matching), and visual aesthetics. Together, these components yield sketches that are both interpretable to humans and aligned with the intended target.

However, our method also reveals several limitations. First, while effective on single-object images, the system struggles with complex scenes involving multiple overlapping objects or fine-grained textures. This suggests a need for hierarchical policies that can reason over stroke groups or semantic regions rather than individual strokes alone. Second, CLIP-based rewards, while powerful for guiding global semantics, are still relatively coarse and can lead to mode collapse or ambiguous strokes without additional structural constraints. Incorporating feedback from intermediate layers of vision-language models or leveraging pretrained scene parsers may improve granularity.

Furthermore, although our user study validates perceptual alignment, the evaluation pipeline could benefit from more structured metrics for stroke quality, coverage, and compositional reasoning. We believe integrating domain-specific constraints—such as stroke order priors or object part annotations—could further enhance both controllability and generalization.

Overall, our work takes a step toward human-aligned sketch generation systems that learn to draw in a goal-directed, interpretable manner. Future directions include scaling to multi-object scenes, exploring hierarchical reinforcement learning, and enabling user-in-the-loop sketch editing through interactive policies.

7 Conclusion

We presented a reinforcement learning framework for generating semantically meaningful and visually coherent vector sketches, inspired by the natural human sketching process. By decomposing the task into stroke placement and refinement, and leveraging a combination of CLIP-based semantic feedback, geometric alignment, and aesthetic regularization, our method produces sketches that align well with input images and are interpretable to human users. While challenges remain in scaling to complex scenes and fine-grained detail, our results highlight the potential of structured RL approaches for controllable, goal-directed sketch synthesis.

8 Team Contributions

- **Group Member 1:** Mia Tang: Responsible for all tasks involvde: algorithm design, implementation, and evaluation.

Changes from Proposal The main shift is moving from training a unified sequential agent to the two-stage separate RL approach. The object stayed the same.

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